PROBLEM 46

Bumpy Taxiway

Original statement · Problems.pdf, p. 31

Resonance & Damping

An airplane is taxiing over a bumpy runway of vertical profile $y_1 = \sin(x)$ with a horizontal velocity c. The vertical motion of the airplane can be modeled as that of a heavy particle P of mass m supported on a landing gear of nominal length L when unloaded. Consider that the landing gear has stiffness constant k and negligible damping.

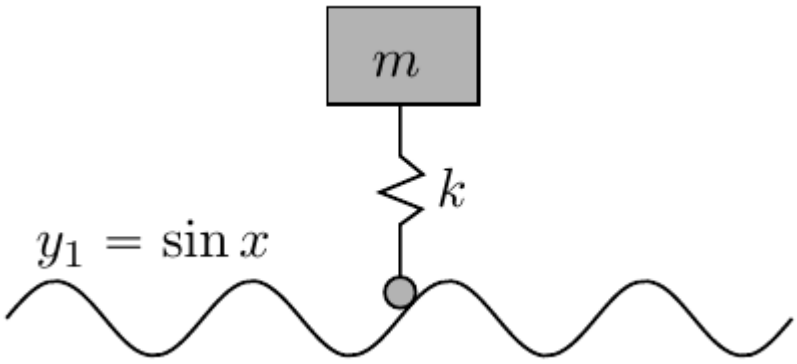


Figure: Mass m on a spring k whose lower end follows the bumpy runway profile y1 = sin x (Problems.pdf, p. 31).

(a) At what taxiing speed is the system in resonance with the bumpy road?

(b) Find the expression of the amplitude of the airplane oscillations when the taxiing velocity c is very high. Hint: the expression depends on c.

(c) Consider now that the landing gear has damping constant λ. How does the results above change? Consider the underdamped, critical, and overdamped cases. What value of λ would you choose for this application and why? If the mass of the aircraft changes (e.g. more or less payload), does this affect your conclusion?

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** the airplane is modelled as a heavy particle P of mass m on a landing gear of stiffness k and nominal (unloaded) length L that rolls at the horizontal speed c on the runway $y_1 = \sin x$. In part (c) a damper of constant λ acts between the fuselage and the wheel.
- **Constraints:** the particle moves only vertically (the horizontal motion at speed c is prescribed and does not enter the vertical dynamics), and the wheel stays on the ground: $y_b(t) = \sin(ct)$ with $x = ct$. This is a prescribed, time-dependent (rheonomic) base motion. The wheel contact is a unilateral constraint, assumed to remain active.
- **Degrees of freedom:** $\mathrm{CDOF} = 1$, with the height y of P.
- **Hypotheses:** the profile is written with unit wavenumber ($1\ \mathrm{m}^{-1}$), so the excitation has angular frequency $\Omega = c$ (rad/s when c is in m/s); gravity acts on m; the gear is linear.
- **Parameters:** m [kg], k [N/m], L [m], c [m/s], g [m/s^2], $\omega_0 = \sqrt{k/m}$ [rad/s], λ [N s/m].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ with Oy vertical upward (constant unit vectors).

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the particle moves along the vertical, and the length of the spring is the distance between P and the ground point under the wheel:

$$\mathbf{r}_P = y\,\mathbf{j}, \qquad \mathbf{a}_P = \ddot{y}\,\mathbf{j}, \qquad y_b(t) = \sin(ct), \qquad \dot{y}_b = c\cos(ct), \qquad \ell_s = y - y_b$$

3.2 Forces

- **Weight:** $-mg\,\mathbf{j}$. **Spring force:** $-k\,(\ell_s - L)\,\mathbf{j}$ (upward when compressed). **Damper (part c):** $-\lambda\,(\dot{y} - \dot{y}_b)\,\mathbf{j}$, proportional to the relative velocity between the fuselage and the wheel. **Inertia forces:** none.

3.3 Equations of Motion

- **Newton's second law along Oy:**

$$m\ddot{y} = -k\,[y - \sin(ct) - L] - mg$$

- The constant weight only shifts the equilibrium. With $y = z + L - mg/k$ (so $\ddot{y} = \ddot{z}$ and $-k(y - L) - mg = -kz$), z measures the motion about the static equilibrium over flat ground:

$$m\ddot{z} = -kz + k\sin(ct) \implies \ddot{z} + \omega_0^2 z = \omega_0^2 \sin(ct)$$

3.4 Integration and Answers

(a) Resonance

Pedagogical Justification: this is a harmonic oscillator forced at the angular frequency $\Omega = c$ by the base motion (Notes Sec. 5.4). Resonance occurs when the forcing frequency equals the natural frequency: the amplitude of the forced response, $A = \omega_0^2/(\omega_0^2 - c^2)$ derived in (b) below, diverges at $c = \omega_0$. There the trial $A\sin(ct)$ fails (it solves the homogeneous equation) and the particular solution is $z_p = -\frac{1}{2}\omega_0 t\cos(\omega_0 t)$ (check: $\ddot{z}_p + \omega_0^2 z_p = \omega_0^2\sin(\omega_0 t)$), which grows linearly in time, as in Problem 45.

$$\boxed{c_{\text{res}} = \omega_0 = \sqrt{\frac{k}{m}}}$$

For a profile of general wavelength Λ the resonant speed would be $c = \omega_0\Lambda/2\pi$ (here $\Lambda = 2\pi$).

(b) Amplitude at very high speed

> **Pedagogical Justification:** the motion is the free oscillation at ω_0 plus the forced response; with negligible damping the free part does not decay and depends only on the initial conditions. The amplitude asked for is that of the forced part, the particular solution $z_p = A\sin(ct)$ for $c \neq \omega_0$ (in units of the bump height, 1 m):

$$\ddot{z}_p = -c^2 A \sin ct \implies \left(\omega_0^2 - c^2\right) A = \omega_0^2 \implies A = \frac{\omega_0^2}{\omega_0^2 - c^2}$$

$$\boxed{|A(c)| = \frac{\omega_0^2}{|\omega_0^2 - c^2|} \xrightarrow[c \gg \omega_0]{} \frac{\omega_0^2}{c^2} = \frac{k}{mc^2}}$$

For very high speed the amplitude decays as $1/c^2$: the mass cannot follow the fast bumps, and z_p is in antiphase with the road ($A < 0$ for $c > \omega_0$).

(c) Landing gear with damping λ

> **Pedagogical Justification:** the damper adds $-\lambda\,(\dot{y} - \dot{y}_b)$. With $a = \lambda/m = 2\zeta\omega_0$ and the damping ratio $\zeta = \lambda/\left(2\sqrt{km}\right)$:

$$\ddot{z} + a\dot{z} + \omega_0^2 z = \omega_0^2 \sin(ct) + ac\cos(ct)$$

- **Transient (homogeneous solution).** The characteristic equation $r^2 + ar + \omega_0^2 = 0$ has the roots $r = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$:
  - Underdamped, $\lambda < 2\sqrt{km}$ ($\zeta < 1$): $z_h = e^{-\zeta\omega_0 t}\left(C_1\cos\omega_d t + C_2\sin\omega_d t\right)$ with $\omega_d = \omega_0\sqrt{1-\zeta^2}$.
  - Critical, $\lambda = 2\sqrt{km}$ ($\zeta = 1$): $z_h = (C_1 + C_2 t)e^{-\omega_0 t}$, the fastest return without oscillation.
  - Overdamped, $\lambda > 2\sqrt{km}$ ($\zeta > 1$): $z_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ with $r_{1,2} < 0$ real, a slower return (the three regimes of Notes Sec. 5.3).

- **Steady state.** The forcing is the imaginary part of $\left(\omega_0^2 + iac\right) e^{ict}$, so with $z_p = \mathrm{Im}\left[He^{ict}\right]$. Substituting $\tilde{z} = He^{ict}$, the chain rule gives $\dot{\tilde{z}} = ic\,\tilde{z}$ and $\ddot{\tilde{z}} = -c^2\tilde{z}$, so the equation becomes $\left(\omega_0^2 - c^2 + iac\right) H = \omega_0^2 + iac$ (Notes Sec. 5.2, complex representation):

$$H = \frac{\omega_0^2 + iac}{\omega_0^2 - c^2 + iac}, \qquad |H|^2 = \frac{\omega_0^4 + a^2c^2}{\left(\omega_0^2 - c^2\right)^2 + a^2c^2}$$

The steady amplitude is $|H|$ and it is always finite. At resonance, $c = \omega_0$: $|H(\omega_0)| = \sqrt{1 + \dfrac{\omega_0^2}{a^2}} = \sqrt{1 + \dfrac{1}{4\zeta^2}}$. For $c \gg \max\left(a, \omega_0^2/a\right)$ (so that the damping term dominates both the numerator and the denominator), $|H| \approx ac/c^2 = a/c = \lambda/(mc)$: it decays only as $1/c$, slower than the undamped $1/c^2$ of part (b).

- **The steady state is the same in the three cases.** The formula $|H|$ does not depend on whether the gear is underdamped, critical or overdamped; only the transient z_h differs, and it dies out in all three.
- **Effect on the results of (a) and (b).** With $r = c/\omega_0$ and $a = 2\zeta\omega_0$ (extension notation; the notes use $\gamma = \lambda/(2m)$, with $\zeta = \gamma/\omega_0$), $|H|^2 = \dfrac{1 + 4\zeta^2 r^2}{\left(1 - r^2\right)^2 + 4\zeta^2 r^2}$. The resonance is no longer infinite: the maximum of $|H|$ moves below ω_0, to $r_p^2 = \dfrac{\sqrt{1 + 8\zeta^2} - 1}{4\zeta^2} < 1$ (with $u = r^2$, $N = 1 + 4\zeta^2 u$ and $D = (1-u)^2 + 4\zeta^2 u$, the quotient rule gives $N'D - ND' = 4\zeta^2\left[(1-u)^2 + 4\zeta^2 u\right] - (1 + 4\zeta^2 u)\left[-2(1-u) + 4\zeta^2\right] = 2(1 - u - 2\zeta^2 u^2)$, so $d|H|^2/du = 0$ means $2\zeta^2 r^4 + r^2 - 1 = 0$), with $|H|_{\max} = 1.47$ for $\zeta = 0.5$ ($r_p = 0.856$) and 1.28 for $\zeta = 0.7$ ($r_p = 0.788$). At $c = \omega_0$ the value is $|H(\omega_0)| = \sqrt{1 + 1/(4\zeta^2)}$, i.e. 1.41 and 1.23.
- **A fixed point of the curve.** At $c = \sqrt{2}\,\omega_0$ ($r^2 = 2$), $|H|^2 = \dfrac{1 + 8\zeta^2}{1 + 8\zeta^2} = 1$ for every ζ: below this speed damping reduces the transmitted amplitude, above it damping increases it. The high-speed decay is only $1/c$.
- **Choice of λ.** Critical damping gives the fastest return of the transient without oscillation, but a larger λ transmits more amplitude above $c = \sqrt{2}\,\omega_0$ and attenuates the bumps at high speed only as $1/c$. A compromise is ζ somewhat below 1 (underdamped): it keeps the resonant peak moderate (1.28 to 1.47 for $\zeta = 0.7$ to 0.5) without overdamping.
- **Effect of the payload.** The mass changes $\omega_0 = \sqrt{k/m}$, so the resonant speed $c_{\text{res}} = \sqrt{k/m}$ is lower for a heavier aircraft, and $\zeta = \lambda/\left(2\sqrt{km}\right) \propto m^{-1/2}$, so a heavier aircraft is less damped. Since λ is fixed hardware, it must be chosen so that ζ stays below 1 for the

lightest payload (to avoid overdamping) and is still large enough to limit the resonant peak for the heaviest one; the lightest payload then has a larger ζ and a higher resonant speed.

PHASE 4

Interpretation & Checks

- **Dimensional check:** the profile is $y_b = h\sin(\kappa c t)$ with $h = 1$ m and $\kappa = 1\ \mathrm{m}^{-1}$ (Hypotheses), so $\Omega = \kappa c$ is in s^{-1}, $[\omega_0^2] = [k/m] = \mathrm{s}^{-2} = [\Omega^2]$, and the forcing $\omega_0^2 h\sin(\Omega t)$ is in $\mathrm{m/s}^2$; $[a] = [\lambda/m] = \mathrm{N\ s\ m^{-1} kg^{-1}} = \mathrm{s}^{-1}$, so $\zeta = a/(2\omega_0)$ is dimensionless.
- **Limits:** $c \to 0$ gives $|A| \to 1$ (the airplane follows the road quasi-statically); $\lambda \to 0$ in $|H|$ recovers $|A|$; for $\zeta \to \infty$ at fixed c, $|H| \to 1$ for every speed (the gear locks and the airplane follows the road); the $|H| \approx \lambda/(mc)$ regime needs $c \gg \max\left(\lambda/m,\ \omega_0^2 m/\lambda\right)$ (otherwise the response still decays as ω_0^2/c^2).
- **Check against the official key:** `Problems.pdf` has no entry for Problem 46 in its solution keys, so the results are verified by independent derivation. Sympy confirms that the undamped particular solution and the damped steady state satisfy their equations with zero residual; numerical integration of the damped equation ($\omega_0 = 1.7$, $a = 0.6$, $c = 2.3$, over 120 s) gives a steady amplitude of 1.156806, equal to $|H| = 1.156806$, and $|H(\omega_0)|^2 = 9.0278 = 1 + \omega_0^2/a^2$.

Source: Problems.pdf